THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 5–8: **Circles, Measuring Space: Perimeter and Area, The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 cm	$r = \sqrt{8^2 + 6^2} = 10$
2	**(d)** 24 cm	half-chord $= \sqrt{15^2 - 9^2} = 12$, so chord $= 2 \times 12 = 24$
3	**(c)** concyclic points	definition
4	**(c)** 25 sq. units	$s = 10$; $\sqrt{(s-a)(s-b)(s-c)(s-d)} = 25$
5	**(a)** 10.5 cm	$r = \frac{C}{2\pi} = \frac{66 \times 7}{44} = 10.5$
6	**(a)** 44 cm	$2\pi r = 2 \times \frac{22}{7} \times 7 = 44$
7	**(b)** $\frac{4}{11}$	4 favourable out of 11 letters
8	**(a)** the Gambler's Fallacy	past outcomes do not affect the next fair toss
9	**(d)** 600	expected number $= P \times$ population
10	**(a)** neither an AP nor a GP	differences 3,5,7; ratios 4, 2.25, …
11	**(b)** −13	$a + (n-1)d = 11 + 8 \times (-3) = -13$
12	**(c)** 32	substitute $n = 12$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$2x + 10 + 3x - 20 = 180 \Rightarrow 5x = 190 \Rightarrow x = 38$; $\angle P = 86^\circ, \angle R = 94^\circ$.	1+1
14	Arc $= \pi r = 22$ cm; diameter $= 14$ cm; perimeter $= 36$ cm.	1+1
15	Perimeter $= 2\pi r$, so the ratio of perimeters equals the ratio of radii: $5 : 4$.	2
16	Primes: 2, 3, 5 ⇒ $\frac{3}{6} = \frac{1}{2}$.	1+1
17	$T_4 = 7,\ T_5 = 13,\ T_6 = 24$.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Distances: $\sqrt{25-9} = 4$ and $\sqrt{25-16} = 3$ (2). Opposite sides: $4 + 3 = 7$ cm (1).	2+1
19	Area $= 42$ cm^2 (Heron, $s = 21$) (2). $\frac{1}{2} \times 7 \times h = 42 \Rightarrow h = 12$ cm (1).	2+1
20	$P = 0.35$; expected $= 0.35 \times 1200 = 420$ (2). A larger sample drawn from different classes represents the whole population better (1).	2+1
21	$500000 + 20000k = 600000 \Rightarrow k = 5$ raises (so he earns ₹6,00,000 in his 6th year).	2+1
22	12, 15, …, 99: $99 = 12 + (n-1)3 \Rightarrow n = 30$ (1). Sum $= \frac{30}{2}(12 + 99) = 1665$ (2).	1+2
	OR: $d = 3$: $t_n = 1 + (n-1)3 = 3n - 2$ (2). Recursive: $t_1 = 1,\ t_n = t_{n-1} + 3$ (1).	2+1
23	The diagonal is $\sqrt{5^2 + 12^2} = 13$; the quadrilateral is made of two right triangles with legs 5 and 12 (1+1): area $= 2 \times \frac{1}{2} \times 5 \times 12 = 60$ square units (1).	1+1+1
	OR: $\angle ACB = 90^\circ$ (angle in a semicircle) (1); $\angle ABC = 180^\circ - 90^\circ - 35^\circ = 55^\circ$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) $\frac{22}{7} \times 196 = 616$ m^2 (1). (ii) Outer radius 21 m; area $\frac{22}{7} \times 441 = 1386$ m^2 (2). (iii) $1386 - 616 = 770$ m^2 (1). (iv) $770 \times 50 =$ ₹38,500 (1).	1+2+1+1
	OR: (i) $36.5 + 0.3 = 36.8$ m (1). (ii) $168.78 + 2\pi(36.8) = 168.78 + 231.22 = 400$ m (2). (iii) Extra radius 1.22 m: $2\pi \times 1.22 \approx 7.67$ m (1). (iv) Each lane's radius increases by the same 1.22 m, so the extra bend distance is the same $2\pi \times 1.22$ for each pair (1).	1+2+1+1
25	(i) $18,\ 13.5,\ 10.125,\ 7.59375,\ 5.695$ (2). (ii) $t_n = 24\left(\frac{3}{4}\right)^n$ (1). (iii) After 7 bounces 3.20 ft; after the 8th 2.40 ft ⇒ 8 bounces (2).	2+1+2

	OR: (i) 5,00,000; 5,20,000; 5,40,000 (1). (ii) $500000 + 7 \times 20000 = ₹6,40,000$ (1). (iii) $500000 + 20000(n-1) = 800000 \Rightarrow n = 16$ (1). (iv) $\frac{10}{2}(2 \cdot 500000 + 9 \cdot 20000) = 5 \times 1180000 = ₹59,00,000$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5–8: **Circles, Measuring Space: Perimeter and Area, The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** 56°	central angle is double
2	**(a)** the diameter	it passes through the centre
3	**(d)** 48 cm	half-chord $= \sqrt{25^2 - 7^2} = 24$, so chord $= 2 \times 24 = 48$
4	**(a)** 1386 cm^2	$\pi r^2 = \frac{22}{7} \times 21^2 = 1386$
5	**(a)** 72 cm^2	area $= \frac{1}{2}bh$
6	**(a)** 11 cm	$\frac{45}{360} \times 2\pi r = 11$
7	**(c)** $\frac{4}{11}$	4 favourable out of 11 letters
8	**(a)** 0	can never happen
9	**(d)** $\frac{1}{3}$	favourable outcomes: 1, 2 ⇒ 2 out of 6
10	**(d)** −28	substitute $n = 6$
11	**(d)** $n = 25$	$4n - 3 = 97 \Rightarrow n = 25$
12	**(c)** 30	3, 6, 30

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$\angle C = 180^\circ - 75^\circ = 105^\circ$; $\angle D = 180^\circ - 110^\circ = 70^\circ$.	1+1
14	$2\pi r = 44 \Rightarrow r = \frac{44 \times 7}{2 \times 22} = 7$ cm.	1+1
15	Sample space $\{HH, HT, TH, TT\}$; favourable: HT, TH ⇒ $P = \frac{2}{4} = \frac{1}{2}$.	1+1
16	Not red = blue: $\frac{6}{10} = \frac{3}{5}$.	1+1
17	$t_1 = 2,\ t_2 = 3,\ t_3 = 6,\ t_4 = 11,\ t_5 = 18$.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Half-chord $= \sqrt{15^2 - 9^2} = 12$; chord $= 24$ cm.	2+1
19	Triangle ABC (the centres and an intersection point) is equilateral, so $\angle CAD = 120^\circ$ (1). Each outer arc is $240^\circ = \frac{2}{3}$ of a circle (1). Perimeter $= 2 \times \frac{2}{3} \times 2\pi r = \frac{8\pi r}{3}$ (1).	1+1+1
20	(i) $\frac{24}{120} = 0.2$ (1). (ii) $\frac{22+21+16}{120} = \frac{59}{120} \approx 0.49$ (1). Theoretical for 5 is $\frac{1}{6} \approx 0.167$; close but not equal (1).	1+1+1
21	$t_{10} = 3 \cdot 2^9 = 1536$ (2). $t_1 = 3,\ t_n = 2t_{n-1}$ (1).	2+1
22	Arcs BCD and BAD together make the whole circle. $\angle BAD = \frac{1}{2}$(reflex $\angle BOD$) and $\angle BCD = \frac{1}{2}\angle BOD$ (2). Sum $= \frac{1}{2} \times 360^\circ = 180^\circ$ (1).	2+1
	OR: The arc not containing D subtends a straight angle 180° at the centre (1). The angle at D is half of this (1): $\angle ADB = \frac{1}{2} \times 180^\circ = 90^\circ$ (1).	1+1+1
23	$s = 88$; $s - a = 63, 49, 36, 28$ (1). Area $= \sqrt{63 \times 49 \times 36 \times 28} = 1764$ cm^2 (2).	1+2
	OR: $s = 2a$; area $= \sqrt{(2a - a)^4} = \sqrt{a^4} = a^2$ (3), which is the known area of the square.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Join D to C and extend to E on arc AFB (1). $\triangle DCB$ is isosceles ($CB = CD$) ⇒ $\angle CBD = \angle CDB$; exterior angle $\angle BCE = 2\angle BDC$ (2). Likewise $\angle ACE = 2\angle CDA$ (1). Add: $\angle BCA = 2(\angle BDC + \angle CDA) = 2\angle BDA$ (1).	1+2+1+1
	OR: Let $AB > DE$ with perpendiculars CF, CG from the centre C. $AC = CD$ (radii). By Pythagoras: $CF^2 + AF^2 = CG^2 + GD^2$ (2). F and G are midpoints, so $AF > GD$ (1) ⇒ $AF^2 > GD^2$ ⇒ $CF^2 < CG^2$ (1) ⇒ $CF < CG$ (1).	2+1+1+1
25	(i) Certain (1). (ii) Impossible (0). (iii) Less likely ($\frac{1}{6}$). (iv) Even chance ($\frac{1}{2}$). (v) Certain (1). (1 mark each)	1+1+1+1+1

	OR: (i) $\frac{1}{13}$ (ii) $\frac{1}{2}$ (iii) $\frac{3}{13}$ (iv) $\frac{26}{52} = \frac{1}{2}$ (v) $\frac{1}{52}$ (1 mark each).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5–8: **Circles, Measuring Space: Perimeter and Area, The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** equal	equal chords ⇔ equal distances
2	**(a)** 8 cm	half-chord $= \sqrt{5^2 - 3^2} = 4$, so chord $= 2 \times 4 = 8$
3	**(d)** 25 cm	$r = \sqrt{7^2 + 24^2} = 25$
4	**(a)** 11 cm	$\frac{90}{360} \times 2\pi r = 11$
5	**(d)** $\frac{\sqrt{3}}{4}a^2$	Heron with $s = \frac{3a}{2}$
6	**(d)** 220 cm	$2\pi r = 2 \times \frac{22}{7} \times 35 = 220$
7	**(b)** $\frac{1}{3}$	favourable outcomes: 5, 6 ⇒ 2 out of 6
8	**(c)** $\frac{2}{11}$	2 favourable out of 11 letters
9	**(a)** 36	multiply the number of outcomes of each experiment
10	**(c)** $n = 113$	$3n - 7 = 332 \Rightarrow n = 113$
11	**(b)** 21	13+8=21
12	**(d)** 81	substitute $n = 20$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	90° — the angle subtended by a diameter at any point of the circle (angle in a semicircle).	1+1
14	Circumference $= \frac{22}{7} \times 56 = 176$ cm; distance $= 100 \times 176 = 17600$ cm $= 176$ m.	1+1
15	(i) Certain (probability 1). (ii) Impossible (probability 0, or essentially so).	1+1
16	$-5, -2, 1, 4, 7$ (1). $t_n = -5 + 3(n-1) = 52 \Rightarrow n = 20$, so 52 is the 20th term (1).	1+1
17	$5n - 3 = 97 \Rightarrow n = 20$ — yes (20th term). $5n - 3 = 172 \Rightarrow n = 35$ — yes (35th term).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Let $OM \perp AB$. In right triangles OMA and OMB: $OA = OB$ (radii), OM common, $\angle OMA = \angle OMB = 90^\circ$ ⇒ congruent (RHS) (2). So $AM = MB$ (1).	2+1
19	Radii a', b', c', d'. Path $a = \pi a'$; path $b + c + d = \pi(b' + c' + d')$ (1). $PQ = 2a' = 2b' + 2c' + 2d' \Rightarrow a' = b' + c' + d'$ (1). Hence the lengths are equal (1).	1+1+1
20	$\frac{5}{500} = \frac{1}{100} = 0.01$.	2+1
21	Last term $= 3 + 19 \times 5 = 98$ (1). Sum $= \frac{3+98}{2} \times 20 = 1010$ (2).	1+2
22	Circle $= \frac{22}{7} \times 21^2 = 1386$ cm² (1+1). Sector $= \frac{60}{360} \times 1386 = 231$ cm² (1).	2+1
	OR: Straights: $2 \times 84.39 = 168.78$ m (1). Two semicircles make one circle: $2\pi r = 2 \times 3.1416 \times 36.5 = 229.34$ m (1). Total ≈ 398.12 m (1).	1+1+1
23	$P(\text{not } E) = 1 - \frac{5}{12} = \frac{7}{12}$ (2). Sum $= \frac{5}{12} + \frac{7}{12} = 1$ ✓ (1).	2+1
	OR: 36 outcomes (1). (i) (2,6),(3,5),(4,4),(5,3),(6,2) ⇒ $\frac{5}{36}$ (1). (ii) (4,6),(5,5),(5,6),(6,4),(6,5),(6,6) ⇒ $\frac{6}{36} = \frac{1}{6}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Cut the right triangle off one side and attach it to the other to form a rectangle of the same base b and height h (2). Rectangle area $= bh$ (1). For a thin parallelogram the foot of the perpendicular lies outside the base: shift triangles repeatedly (equal congruent triangles CDD', BAA') until the construction works (2).	2+1+2
	OR: (i) $s = a + b$; $\sqrt{b \cdot a \cdot b \cdot a} = ab$ ✓ (2). (ii) $s = 15$; area $= \sqrt{(15-6)(15-14)(15-5)(15-5)} = \sqrt{900} = 30$ (2). Height $= \sqrt{25 - 16} = 3$; $\frac{1}{2}(6 + 14) \times 3 = 30$ ✓ (1).	2+2+1
25	(i) Write the sum forwards and backwards and add: $2S = n(n+1)$ (2). (ii) $S_{20} = 210$, $S_{50} = 1275$ (1). (iii) $S_{58} - S_{24} = 1711 - 300 = 1411$ (2).	2+1+2

	OR: (i) $1, 2, 3, 5, 8, 13, 21, 34, 55, 89$ (2). (ii) $1 + 2 + 3 + 5 + 8 + 13 + 21 + 34 = 87 = V_{10} - 2$ (2). (iii) By Virahānka, 7th century CE, in connection with Prakrit metre (1).	2+2+1

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